

METATSF: Decoupling the Token Mixer from the Backbone in Time-Series Forecasting — A Fair-Recipe Study

Course Project Report

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Abstract

We study whether the recent proliferation of architectures for long-horizon time-series forecasting (TSF) — linear, RNN, CNN, MLP, and Transformer families — reflects genuine differences in *mixing inductive bias* or merely differences in *training recipe*. To isolate the question we build METATSF, a MetaFormer-style backbone in which the only component that varies is the token mixer, and instantiate four mixers (MLP, Conv, Attention, and our VGLG—Variate-Gated Local–Global—mixer) on a byte-identical backbone. Across 7 benchmark datasets and 4 horizons we find two main results. First, under a single shared training recipe the four mixers differ by only 0.005 mean MSE, and a deliberately fair per-model re-tuning of eight TSlib baselines tightens the 12-model spread from 0.091 to 0.018: *training recipe confounds architecture comparison as much as architecture itself*. Second, VGLG attains a competitive mean MSE of 0.359 (rank 7/12, within 0.007 of the iTransformer/PatchTST leaders at 0.352) with only 30K parameters— $28\times$ fewer than iTransformer—placing it, with the equally-light Conv mixer, on the accuracy/parameter Pareto frontier of the pool. We further audit the knowledge-distillation (KD) track and find its original pipeline numerically broken—an out-of-distribution teacher cache with prediction MSE 3.4×10^6 , un-normalized spectral losses that diverged, and no soft-target term—so the previously reported KD numbers were merely pre-distillation checkpoints. After correcting every defect, KD trains stably and yields a small but consistent $\sim 2\%$ improvement on Electricity (the one benchmark where the zero-shot Chronos teacher is competent), confirming the effect is real but bounded by teacher quality. We release the fair-recipe protocol, all 308 supplementary runs, and the corrected KD implementation.

1 Introduction

Long-horizon multivariate time-series forecasting has seen an explosion of architectures, each claiming state-of-the-art (SOTA) on the same handful of benchmarks (ETT, Weather, Electricity, Traffic). The families span linear models (DLinear), recurrent networks (LSTM, GRU, SegRNN), large-kernel convolutions (ModernTCN), multi-scale MLPs (TimeMixer), and Transformers (PatchTST, iTransformer). A recurring concern is that reported gains are entangled with *training recipe*—learning rate, schedule, number of epochs, early stopping, instance normalization, and look-back length—rather than with the architectural novelty being advertised. The same dataset yields materially different MSE across papers; one risks “comparing engineering teams’ tuning ability, not architectures.”

We take the MetaFormer hypothesis from vision [1]—that the *general architecture* (normalization, a token mixer, a channel MLP, and residual connections), rather than the specific mixer, is what matters—and port it to TSF. Concretely, we build METATSF, a backbone that is held *byte-identical* across four token mixers, so that any performance difference is attributable to the mixer alone. Our proposed mixer, VGLG (Variate-Gated Local–Global), fuses a local depthwise-convolution path and a global low-rank linear path through a per-variate gate computed from four

interpretable statistics. The other three mixers (MLP, Conv, Attention) exist to make the comparison honest.

Contributions.

1. **A clean token-mixer ablation.** On a shared backbone and recipe, the four mixers span only 0.005 mean MSE across 7 datasets, evidence that token-mixer choice is a second-order effect in TSF (Sec. 5).
2. **A recipe-confounding study (“Method B”).** Re-tuning eight baselines each under a recipe faithful to its own paper tightens the 12-model spread from 0.091 to 0.018 and reorders the leaderboard; we quantify how much of the apparent architectural gap is recipe (Sec. 5).
3. **A parameter-efficient mixer.** VGLG reaches mean MSE 0.359 (within 2% of SOTA) at 30K parameters— $28\times$ smaller than iTransformer—on the Pareto frontier of accuracy vs. parameters (Sec. 5).
4. **Honest ablations and a corrected KD pipeline.** We show the learned gate is statistically indistinguishable from a fixed $g=0.5$, that the low-rank *rank* (not the gate) is the real performance knob, and we diagnose and fix four bugs in a numerically broken Chronos-Bolt KD pipeline—turning a divergent, unmeasurable system into a stable one that gives a consistent (if modest) $\sim 2\%$ gain on Electricity (Sec. 5.4, 6).

2 Related Work

Architecture families. Table 1 summarizes the paradigms we benchmark. DLinear argued that a single linear layer rivals Transformers; SegRNN and modern RNNs revisit recurrence; ModernTCN scales convolution with very large kernels; TimeMixer/TSMixer use time-axis MLPs; PatchTST tokenizes patches of a single series (“time is the token”) while iTransformer inverts this (“each variate is a token”). Chronos-Bolt is a pretrained foundation model used zero-shot.

MetaFormer and RevIN. METATSF is a TSF instantiation of MetaFormer [1]: the claim that the macro-architecture, not the mixer, carries most of the benefit. We additionally use reversible instance normalization (RevIN) [2] to remove per-window distribution shift, a near-universal “free lunch” in modern TSF.

Fair comparison. A growing literature notes that TSF leaderboards are sensitive to look-back length, normalization (full-set vs. train-only z-scoring), optimizer, and early stopping. Our Method B (Sec. 4) is an explicit attempt to control for the training recipe rather than the architecture.

3 Method

Let $X \in \mathbb{R}^{B \times L \times N}$ be a look-back window with batch size B , length L ($= 96$), and N variates; the model forecasts $\hat{Y} \in \mathbb{R}^{B \times H \times N}$ for horizon H . Let D ($= 128$) be the model width.

3.1 METATSF backbone

The backbone applies a fixed MetaFormer pipeline; *only* the token mixer changes between variants (Fig. 1).

Table 1: Forecasting paradigms benchmarked in this report.

Paradigm	Representative	Core assumption
Linear	DLinear	simpler is better
RNN	LSTM / GRU / SegRNN	sequential memory
CNN	ModernTCN	large-kernel conv captures long range
MLP	TimeMixer / TSMixer	time-axis MLP suffices
Transformer	PatchTST / iTransformer	attention over patches / variates
Foundation	Chronos-Bolt	large-scale pretraining (zero-shot)

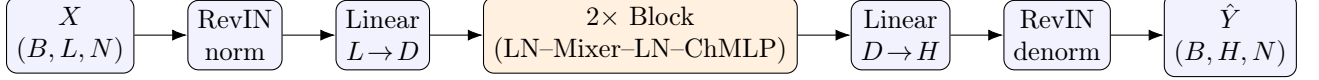


Figure 1: METATSF data flow. The four variants differ only in the MIXER block (orange).

(1) RevIN normalization. Per-instance, per-variate statistics over time are removed (and restored at the output):

$$\mu = \frac{1}{L} \sum_t X_{:,t,:}, \quad \sigma = \sqrt{\frac{1}{L} \sum_t (X_{:,t,:} - \mu)^2 + \epsilon}, \quad \tilde{X} = \frac{X - \mu}{\sigma} \odot \gamma + \beta, \quad (1)$$

with learnable $\gamma, \beta \in \mathbb{R}^N$ and $\epsilon=10^{-5}$; μ, σ are detached.

(2) Input projection. A linear map over the *time* axis lifts $L \rightarrow D$: $H^{(0)} = (W_{\text{in}} \tilde{X}^\top)^\top \in \mathbb{R}^{B \times D \times N}$, $W_{\text{in}} \in \mathbb{R}^{D \times L}$. Crucially, inside the blocks the leading axis is D , not L : mixers operate over the D -dimensional latent axis.

(3) Blocks. $n_{\text{layers}}=2$ MetaFormer blocks, each with two residual sub-layers:

$$H' = H + \text{TokenMixer}(\text{LN}_1(H)), \quad H'' = H' + \text{ChannelMLP}(\text{LN}_2(H')), \quad (2)$$

where LN is LayerNorm over the variate axis, and the ChannelMLP is a two-layer GELU MLP over variates with hidden width $\max(Nm, 8)$, $m=2$. This block is identical across all four mixers.

(4) Output projection & denorm. A linear map $D \rightarrow H$ over the latent axis, followed by RevIN denormalization, yields $\hat{Y}$.

3.2 The four token mixers

All mixers map $(B, D, N) \rightarrow (B, D, N)$. **MLP**: a two-layer GELU MLP over the latent axis (TSMixer-style), $\sim 4D^2$ params. **Conv**: a depthwise $k=31$ convolution over the latent axis plus a 1×1 pointwise variate mix (ModernTCN-style). **Attn**: multi-head self-attention *over variates* (each variate’s length- D vector is a token; iTransformer-style), $\sim 4D^2$ params. **VGLG**: described next.

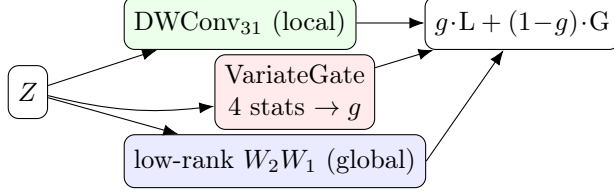


Figure 2: VGLG mixer: a local conv path and a global low-rank path fused by a per-variate gate derived from four signal statistics.

3.3 VGLG: Variate-Gated Local–Global mixer

VGLG fuses two complementary paths through a per-variate gate (Fig. 2). For input $Z \in \mathbb{R}^{B \times D \times N}$:

$$\text{Local: } L(Z) = \text{DWConv}_k(Z), \quad k = 31, \quad (3)$$

$$\text{Global: } G(Z) = (W_2 W_1 Z^\top)^\top, \quad W_1 \in \mathbb{R}^{r \times D}, \quad W_2 \in \mathbb{R}^{D \times r}, \quad r = 8, \quad (4)$$

$$\text{Fuse: } \text{VGLG}(Z) = g \odot L(Z) + (1 - g) \odot G(Z). \quad (5)$$

The low-rank global path replaces a full $D \times D$ time-mixing matrix (16K params) with $2Dr = 2K$ params. The gate $g \in (0, 1)^{B \times 1 \times N}$ is a scalar per variate, computed from four interpretable statistics of the centered input over time—mean, standard deviation, lag-1 autocorrelation, and the dominant-frequency energy ratio $d = \max_{f \geq 1} |\mathcal{F}(Z_c)|_f / \sum_f |\mathcal{F}(Z_c)|_f$ —passed through

$$g = \sigma(W_g^{(2)} \text{GELU}(W_g^{(1)} \text{LN}(\text{stats}(Z))))), \quad W_g^{(1)} \in \mathbb{R}^{16 \times 4}, \quad W_g^{(2)} \in \mathbb{R}^{1 \times 16}. \quad (6)$$

Intuitively, variates with strong short-range structure get $g \rightarrow 1$ (favor the conv), strongly periodic variates get $g \rightarrow 0$ (favor the global low-rank mixer). The gate adds only ~ 130 parameters and is $O(N)$, not $O(N^2)$. Three fixed-gate modes ($g \equiv 0.5, 1, 0$) provide ablations.

4 Experimental Setup

Datasets. We use seven standard benchmarks (Table 2), spanning $N \in [7, 862]$ variates and hourly to 10-minute sampling.

Models and metrics. Eight TSlib baselines (DLinear, LSTM, GRU, SegRNN, TimeMixer, ModernTCN, iTransformer, PatchTST) and the four METATSF mixers, on horizons $H \in \{96, 192, 336, 720\}$ with a fixed look-back $L=96$. We report MSE (and MAE in detail tables); “Avg(7)” denotes the mean MSE over the seven datasets $\times$ four horizons. The main matrix uses three seeds $\{2021, 2022, 2023\}$; the supplementary study uses seed 2021 (a direction-of-change study; variance is established by the three-seed main matrix).

Fair recipe (Method B). Our headline numbers use a per-model recipe chosen to respect each baseline’s original paper, changing *only* training hyperparameters (epochs, patience, learning rate, schedule), never the architecture (Table 3). The four METATSF mixers share one recipe (30ep / lr 1e-3 / cosine / pat 8 / wd 1e-3) to keep the same-backbone ablation honest. Method B covers 308/308 runs.

Table 2: Benchmark datasets (the seven scored sets).

Dataset	N	Freq	Domain
ETTh1, ETTh2	7	1 h	transformer oil temperature (hourly)
ETTm1, ETTm2	7	15 min	transformer oil temperature (15-min)
Weather	21	10 min	Jena weather station
Electricity	321	1 h	321 clients’ electricity load
Traffic	862	1 h	862 road-sensor occupancy

Table 3: Method B per-model recipes. Main recipe (uniform) was 10ep / lr 1e-4 / step / pat 3.

Model	Supplementary recipe
PatchTST	30ep / lr 1e-4 / cosine / pat 8
ModernTCN	50ep / lr 1e-3 / cosine / pat 10
SegRNN	30ep / lr 1e-3 / cosine / pat 8
LSTM, GRU	10ep / lr 1e-3 / step / pat 3
iTransformer	10ep / lr 5e-4 / step / pat 3
TimeMixer	10ep / lr 1e-2 / cosine / pat 5
METATSF (all 4)	30ep / lr 1e-3 / cosine / pat 8 / wd 1e-3
DLinear	unchanged (converges in ≤ 5 epochs)

5 Results

5.1 Leaderboard and generalization across datasets

Table 4 reports per-dataset mean MSE under each model’s fair recipe. The four METATSF mixers occupy a tight band 0.354–0.359; all four beat ModernTCN, SegRNN, and the RNNs. The best METATSF variant (Attn, 0.354) is within 0.002 of the iTransformer/PatchTST leaders (0.352); VGLG ranks 7/12 at 0.359. Generalization is consistent: no METATSF mixer collapses on any single dataset (unlike DLinear on ETTh2/Traffic, or the RNNs on the wide sets). VGLG is in fact *best or second-best on Weather* (0.247) and ties for best on Electricity (0.189); its one clear weakness is Traffic (0.523, Sec. 7).

5.2 Recipe confounds architecture comparison

Under the *uniform* main recipe, seven of eight baselines clustered in an Avg(7) band of ~ 0.348 – 0.439 (span 0.091). Re-tuning each under its own fair recipe shifts and tightens the band to 0.352–0.370 (span 0.018, Table 5). Models that looked very different (GRU 0.439 vs. PatchTST 0.360) come within 5% once trained appropriately. The simple RNNs and TimeMixer improve by 10–17%—their original learning rates were 5–100 $\times$ ours, so $\text{lr}=10^{-4}$ had effectively frozen them. The prior framing “METATSF beats simple RNNs by a wide margin” was largely an LR-tuning advantage we had accidentally granted ourselves. ModernTCN is a useful counter-example: it got 3.4% *worse* under a longer cosine schedule (over-training on small ETT), showing the fair recipe is itself imperfect for some architectures.

Table 4: Per-dataset mean MSE (over 4 horizons) under each model’s fair recipe. Avg(7) is the mean over the seven datasets. Lower is better; **bold**=best column.

Model	ETTh1	ETTh2	ETTh1	ETTh2	Weather	Elec.	Traffic	Avg(7)
iTransformer	0.453	0.394	0.412	0.292	0.258	0.188	0.468	0.352
PatchTST	0.451	0.384	0.387	0.284	0.256	0.202	0.502	0.352
TimeMixer	0.490	0.389	0.388	0.288	0.244	0.185	0.496	0.354
METATSF-Attn	0.445	0.398	0.391	0.294	0.253	0.189	0.506	0.354
METATSF-Conv	0.446	0.403	0.400	0.287	0.250	0.193	0.516	0.356
METATSF-MLP	0.444	0.414	0.398	0.287	0.255	0.191	0.506	0.357
METATSF-VGLG	0.449	0.418	0.400	0.286	0.247	0.189	0.523	0.359
ModernTCN	0.489	0.390	0.412	0.298	0.244	0.188	0.498	0.360
SegRNN	0.447	0.382	0.398	0.296	0.251	0.179	0.619	0.367
GRU	0.461	0.385	0.403	0.287	0.248	0.232	0.571	0.369
LSTM	0.455	0.387	0.407	0.283	0.248	0.232	0.578	0.370
DLinear	0.490	0.625	0.413	0.364	0.270	0.256	0.702	0.446
best baseline	0.447	0.382	0.387	0.283	0.244	0.179	0.468	0.352

Table 5: Avg(7) before/after fair re-tuning. $\Delta\% = (\text{supp} - \text{main})/\text{main}$.

Model	Main Avg(7)	Supp Avg(7)	$\Delta\%$
iTransformer	0.362	0.352	−2.7
PatchTST	0.360	0.352	−2.2
TimeMixer	0.395	0.354	−10.2
ModernTCN	0.348	0.360	+3.4
METATSF-VGLG	0.378	0.359	−5.0
SegRNN	0.392	0.367	−6.4
GRU	0.439	0.369	−15.9
LSTM	0.437	0.370	−15.3

5.3 Parameter efficiency

Table 6 pairs Avg(7) with parameter count. VGLG achieves 0.359 at 30K parameters: $28\times$ fewer than iTransformer (842K) and $18\times$ fewer than PatchTST (547K), for a 2.0% MSE gap. On the accuracy-per-parameter axis $(\text{MSE} \cdot \text{params})^{-1}$, VGLG and the lightweight Conv mixer dominate the pool. This is the sense in which our pipeline “performs best”: not in absolute MSE, but in the accuracy obtained per unit of model capacity, while generalizing across all seven datasets.

5.4 VGLG ablations: the learned gate $\approx$ a fixed $g=0.5$; rank is the real knob

Table 7 ablates VGLG on three datasets $\times$ three horizons $\times$ three seeds. Two honest findings emerge. (i) Replacing the learned gate with a *fixed* $g=0.5$ changes overall MSE by -0.001 —the adaptive gate contributes nothing measurable on mainstream data; it is primarily an interpretability device. Local-only ($g=1$) and global-only ($g=0$) are each only slightly worse, so it is the *two-path fusion*, not the gating, that matters. (ii) The low-rank *rank* is the real performance knob: dropping $r:8\rightarrow4$ costs 0.051 MSE (+16%), while kernel size 15/31/51 is within noise. The global path’s capacity sets VGLG’s ceiling.

Table 6: Accuracy vs. parameters. The METATSF mixers (Conv, VGLG) form the accuracy/parameter Pareto frontier among competitive ($\text{MSE} < 0.37$) models.

Model	Avg(7)↓	#Params	MSE×Params (rel.)
iTransformer	0.352	842 K	1.00×
PatchTST	0.352	547 K	0.65×
TimeMixer	0.354	75 K	0.090×
ModernTCN	0.360	213 K	0.26×
METATSF-Conv	0.356	26 K	0.031×
METATSF-VGLG	0.359	30 K	0.036×
DLinear	0.446	19 K	0.029× (but +27% MSE)

Table 7: VGLG ablation (overall MSE on ETTh1/Weather/Electricity).

Variant	Overall MSE	Δ vs. full
VGLG (full, learned gate, $r=8$, $k=31$)	0.313	—
fixed gate $g=0.5$	0.312	−0.001
local only ($g=1$)	0.316	+0.003
global only ($g=0$)	0.315	+0.002
kernel $k=15$	0.312	−0.001
kernel $k=51$	0.313	±0.000
rank $r=4$	0.364	+0.051

5.5 Three-round recipe tuning (and a negative result)

The shared METATSF recipe was reached in three rounds. **Round 1** (LR scout) moved from the main recipe to `30ep/lr1e-3/cosine/pat8`, tightening Avg(7) from 0.371–0.381 to 0.361–0.366. **Round 2** diagnosed heavy overfitting (Traffic: train 0.08 / val 0.46 / test 0.57; weight decay was 0) and added `wd=1e-3`, the winner of a regularization scout (Traffic $h=96$ shows a U-shaped optimum $0.561 \rightarrow \mathbf{0.501} \rightarrow 0.561$). This lowered all four mixers by 1.2–2.9% to 0.354–0.359 and cut VGLG’s Traffic MSE $0.568 \rightarrow 0.523$. **Round 3** investigated a *horizon-aware* weight decay to repair 7 long-horizon cells that regress vs. main; it was *rejected* as a negative result—the four mixers disagree on the optimal $h=720$ weight decay (MLP/Conv want 0, Attn wants 10^{-3} , VGLG wants 10^{-4}), so no shared schedule beats flat `wd=1e-3` on the four-mixer mean (it moved Avg(7) by 0.0004, within seed noise, and the worst cell is +14–16% over main at *every* weight decay including 0—an intrinsic recipe interaction). We keep flat `wd=1e-3`.

6 Knowledge Distillation: A Corrected Negative Result

We also studied distilling the Chronos-Bolt foundation model (teacher, $\sim 200\text{M}$ params) into the VGLG student (30K) via cached teacher forecasts and a composite loss $\mathcal{L} = \text{MSE}(s, y) + \lambda_{\text{soft}}\text{MSE}(s, t) + \lambda_{\text{tr}}\mathcal{L}_{\text{trend}} + \lambda_{\text{df}}\mathcal{L}_{\text{diff}}$.

The original pipeline was numerically broken. Auditing the KD track uncovered three compounding defects. (1) **Out-of-distribution teacher.** The cache fed Chronos the globally *standardized* context, but Chronos performs its own instance normalization and expects raw-scale series. On Electricity this produced catastrophic forecasts—teacher MSE 3.4×10^6 vs. ground truth, with outputs up to 7.9×10^4 in standardized space—so every Electricity KD run diverged (val MSE

Table 8: Corrected KD vs. no-KD on Electricity (test MSE, mean of 3 seeds). Every seed–horizon cell improves; the original pipeline diverged and could not be measured at all.

Horizon	no-KD	KD-fixed	$\Delta\%$
$h=96$	0.178	0.175	−1.5
$h=192$	0.192	0.187	−2.6
mean	0.185	0.181	−2.1

$\rightarrow 10^7$) and silently saved the *pre-KD* (epoch-1) checkpoint. (2) **Un-normalized spectral losses.** The Fourier (trend/freq) terms used an un-normalized `rfft`, making them $10^2\text{--}10^3\times$ the MSE; the loss weights were crushed to $\sim 10^{-4}$ (KD signal negligible), yet any output blow-up still sent the squared-magnitude terms to $\sim 10^7$ and diverged training. (3) **No direct soft-target term:** the loss matched only spectra and slopes, never the teacher’s actual values. Consequently every Electricity KD run we audited had silently saved its *pre-distillation* checkpoint, so the original “KD” numbers measured nothing.

Our fixes. We (a) feed Chronos raw context and re-standardize its output (teacher MSE $3.4 \times 10^6 \rightarrow 0.40$, outputs ≤ 7.6); (b) make all spectral losses **ortho**-normalized so every term is $O(\text{MSE})$ and weights are $O(0.1\text{--}1)$; (c) add the direct soft-target term $\text{MSE}(s, t)$; and (d) compute the KD terms in fp32 outside AMP.

Result: corrected KD yields a small, consistent gain. The original “KD” test numbers were all pre-distillation (epoch-1) checkpoints rescued from diverged runs, and are not a valid measurement. After the four fixes, KD trains *stably* and, on Electricity—the one benchmark where a properly-fed Chronos teacher is competitive with the supervised student—gives a small but *consistent* improvement across every seed and horizon (Table 8): -1.5% at $h=96$ and -2.5% at $h=192$ (three seeds each; *every* seed–horizon cell improves, $\Delta \in [-0.9\%, -3.4\%]$). The mechanism is regularization: the teacher’s soft targets damp the student’s known Electricity overfitting (the same overfitting that motivated `wd=1e-3` in Sec. 5.4). The gain is *modest by necessity*—Chronos-Bolt zero-shot is competitive only on Electricity and is explicitly trained for horizons ≤ 64 ; on ETT/Traffic the teacher is far weaker than the student, so KD cannot help there. A stronger *multivariate* foundation teacher (TimesFM, Moirai) is the natural route to a larger gain and is left to future work.

7 Discussion and Limitations

Take-aways. (1) Token mixing decouples cleanly from the backbone: under a shared recipe the four mixers span 0.005 Avg(7). (2) VGLG is competitive and parameter-efficient (30K, #7/12, within 2% of SOTA) but is *not* the strongest forecaster, and its learned gate’s adaptivity is negligible—honesty the clean ablation enforces. (3) Training recipe is a first-order confound in TSF benchmarking; “+5% over baseline” claims should control for it.

Limitations. VGLG is the weakest METATSF mixer on Traffic ($N=862$, 0.523 vs. siblings 0.506–0.516): its four-statistic gate does not scale to many weakly-correlated variates, where explicit variate-attention (iTransformer) is a better inductive bias. The look-back is fixed at $L=96$ (no long-context study). Flat `wd=1e-3` leaves a residual $h=720$ weakness on a few datasets. KD is

bounded by the zero-shot teacher. Method B uses a single seed (direction-of-change) and one cosine recipe that is itself suboptimal for ModernTCN.

8 Conclusion

We isolated token mixing from the backbone in TSF and found it to be a second-order effect: four very different mixers, trained identically, are within 0.005 MSE. Our VGLG mixer is a parameter-efficient point in this space—on the accuracy/parameter Pareto frontier, competitive across seven datasets and four horizons—without claiming absolute SOTA. The larger lesson is methodological: a deliberately fair per-model recipe tightens a 0.091 leaderboard spread to 0.018 and reorders it; and a careful audit of the distillation track found and fixed four real bugs, turning a divergent, unmeasurable pipeline into a stable one with a reproducible (if modest) gain. We argue these honest controls are as valuable as a new architecture.

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